

ELECTIVE COURSE-8.3

UEE721: NONLINEAR AND DIGITAL CONTROL SYSTEMS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: To learn the methods for analyzing the behavior of nonlinear control systems and the design of control systems.				
Syllabus Nonlinear control systems: Introduction to Nonlinear System and their properties, behavior of nonlinear system, classification of nonlinearities, common physical nonlinearities: saturation, friction, backlash, dead -zone, relay, on-off nonlinearity, nonlinear spring, concept of limit cycle, jump resonance, phase-plane method, singular points, stability of nonlinear system, construction of phase trajectories, describing functions approach, stability analysis by describing function method, Lyapunov's stability analysis, Lyapunov's stability criterion, direct method of Lyapunov and the linear systems, method of construction of Lyapunov functions for nonlinear systems. z-plane analysis of discrete-time control systems: Introduction, Impulse sampling and data hold, Reconstructing original signal from sampled signals, concept of pulse transfer function, Realization of digital controllers. Design of discrete-time control systems: Introduction, Stability analysis of closed-loop systems in the z-plane, Transient and steady state response analysis, Design based on the root-locus method, Design based on the frequency-response method. State-space analysis: Introduction, State-space representations of discrete-time systems, Solving discrete-time state-space equations, Pulse transfer function matrix, Discretization of continuous time state space equations, Lyapunov stability analysis, Controllability and Observability, Design via pole placement, State observer design.				
Course Learning Outcomes (CLOs) The students will be able to: 1. Demonstrate non-linear system behaviour by phase plane and describing function methods. 2. Perform the stability analysis nonlinear systems by Lyapunov method 3. Derive discrete-time mathematical models in both time domain (difference equations, state equations) and z-domain (transfer function using z-transform). 4. Predict and analyse transient and steady-state responses and stability linear, time-invariant, discrete-time control systems.				
Text Books 1. Bandyopadhyay, M.N., Control Engineering: Theory and Practice, Prentice-Hall of India Private Limited (2003).				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

2. Khalil, H. K., & Grizzle, J. W. *Nonlinear systems* (Vol. 3). Upper Saddle River, NJ: Prentice hall (2002).
3. Ku, B. C. *Digital control systems*. HRW (1980).

Reference Books

1. Slotine & Li, *Applied Non-Linear Control*, Englewood Cliffs, NJ: Prentice-Hall, (1991).
2. Ogata, K., *Discrete-time Control Systems*, Pearson Education (2005).

Evaluation Scheme:

S N	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessional (Assignments/Projects/Tutorials/Quizzes)	25

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